1. Strategic Problem Statement

1.1. Business Context

Scenario. City Hotel Lisbon sells perishable room inventory ahead of arrival, largely through travel agents and operators. Until the guest arrives, revenue is only expected, not secured.

Strategic issue. Revenue-management research identifies booking cancellations as a core source of demand uncertainty: they limit the accuracy of forecasts, and the usual responses, rigid cancellation policies and overbooking, can themselves damage revenue and reputation (Antonio et al., 2017). For City Hotel Lisbon, this means expected revenue and room plans cannot be fully relied on, and blanket policies are an imperfect fix. Cancellation is therefore a strategic revenue-risk problem, not just an operational event. Its scale is quantified in Section 1.2.

Cross-domain impact. One cancellation touches several domains at once, and we assume (to be validated with hotel stakeholders) that each is managed separately today, so no single view supports a joint decision.

Domain	Assumed current state (to validate)	Gap the project addresses
Distribution / Marketing	Measured: cancellation risk differs sharply by channel and segment (17.3%–68.9%). Assumed: it is not yet used in booking management.	Channel and segment risk guides booking management.
Customer Experience	Assumed: pre-arrival confirmations are manual and not prioritised by risk (simulated in the CX table as about 30% of bookings contacted).	Routine confirmation for all bookings; enhanced, prioritised contact for high-risk bookings.
Finance	Assumed: commission, handling cost and revenue at risk are not linked (the extract has no cost data).	Net value of actions accounts for cost.
Operations	Assumed: resale of freed rooms is not guided by cancellation risk. Simulated resale covers 50.7% of cancelled room-nights (no-shows excluded): a scenario output, not an observed rate.	Resale and capacity decisions are guided by cancellation risk.

Note: entries marked “Assumed” are scenario assumptions for this case, not findings from the dataset; they must be validated with hotel stakeholders. Entries marked “Measured” come from the booking extract.

Transformation opportunity. Integrating booking, customer, cost and inventory data into one decision-support system can shift the hotel from reactive handling to risk-based, proactive booking management, optimizing revenue and room capacity.

1.2. Business Pain Points

Baseline: City Hotel Lisbon, 79,330 bookings, July 2015 – August 2017 (Antonio, Almeida & Nunes, 2019).

Pain point	Supporting data	Why it matters
Limited visibility into cancellation risk	41.7% of bookings (about 33,000) were cancelled or no-show. 12.0% of cancellations (status Canceled; 14.5% if no-shows count as 0-day notice) occur within 7 days of arrival.	Late cancellations leave the least time to resell the room. Early warning is the only way to act before that window closes.
Inefficient allocation of follow-up resources	Travel agents and operators handle 87% of bookings. Cancellation rate ranges from 17.3% (Direct) to 68.9% (Groups). Assumed (to validate): contact with at-risk guests is not systematic.	Risk varies widely across segments, so uniform follow-up wastes effort on safe bookings and misses risky ones. Hospitality churn research on repeat-customer retention supports targeting high-risk customers rather than contacting everyone (van Leeuwen et al., 2025); its transfer to booking cancellation is an assumption to test in a pilot.
Limited ability to optimize revenue and room capacity	€10.9M of €25.3M gross booked room revenue (about 43%) is tied to cancelled bookings. Assumed: freed rooms are not resold based on risk, and commission and handling cost are not linked to risk.	The hotel cannot tell which bookings to act on, or how to adjust follow-up, pricing, capacity allocation and resale so that revenue and occupancy are optimized net of intervention cost. Revenue-management studies show that cancellation-aware pricing, allocation and overbooking decisions can raise profit in simulation (Dong & Ling, 2015; Pimentel et al., 2019).

Note: €10.9M is gross booking value exposed to cancellation, not realized loss. Some of it is resold or recovered through deposits. On one contribution basis (net of commission and room cost, minus the contribution of simulated resale), the simulated net loss proxy is about €3.8M; this is a scenario output, not a measured loss.

1.3. Stakeholders

Group	Role	Interest
General Manager, Revenue Manager	Decision-makers	Set priorities, resources and booking-management strategy; optimize revenue and capacity
Reservation Team	Operational users	Use risk information to follow up with guests and update room availability

Group	Role	Interest
Customer Service / CX	Operational users	Design and send guest messages; own opt-outs and CX KPIs
Finance	Decision support	Own cost and contribution assumptions; judge net value and ROI of interventions
Distribution / Sales	Operational users	Manage agents, tour operators and group-block contracts (release dates, deposits)
Data / Analytics Team	System owners	Build, calibrate and monitor the model; maintain the data pipeline and KPI reporting
Guests	Affected party	Experience the interventions; not direct system users

1.4. Decisions to Improve

Decision	Business question	Supported by
Risk identification	Which bookings are more likely to be cancelled?	Machine Learning
Resource prioritization	Which bookings should staff handle first when resources are limited?	Risk score + value proxy
Intervention selection	Should the hotel confirm, send a reminder or contact the guest directly?	Business rules
Revenue and capacity optimization	Which follow-up, pricing, allocation and resale actions improve revenue and occupancy at a justified cost?	Rule-based analysis + ROI analysis

Only the first decision is addressed directly by ML. The others need business rules, operational data and impact evaluation.

1.5. Strategic Objectives

Each objective follows from a pain point and a decision, and is backed by evidence.

	Objective	Addresses pain	Improves decision	Evidence basis
1	Identify cancellation risk early: score every booking at entry and, as a secondary check, bookings still active at a fixed horizon before arrival	Limited visibility	Risk identification	41.7% cancellation rate; 12.0% of cancellations (Canceled only) within 7 days

	Objective	Addresses pain	Improves decision	Evidence basis
2	Allocate follow-up effort to the highest-risk, highest-value bookings	Inefficient allocation	Resource prioritization; intervention selection	17.3%–68.9% segment gap; risk-based targeting in churn research (van Leeuwen et al., 2025), used as motivation only
3	Optimize revenue and room capacity by supporting follow-up, pricing, allocation and resale decisions, net of intervention cost	Limited ability to optimize revenue and capacity	Revenue and capacity optimization	€10.9M exposure; profit gains in simulated revenue-management studies (Dong & Ling, 2015; Cha, 2026)

Scope of Objective 1: a retrospective prototype built on a public PMS extract with near-arrival values. It demonstrates the approach but does not prove the performance of a live booking-time system.

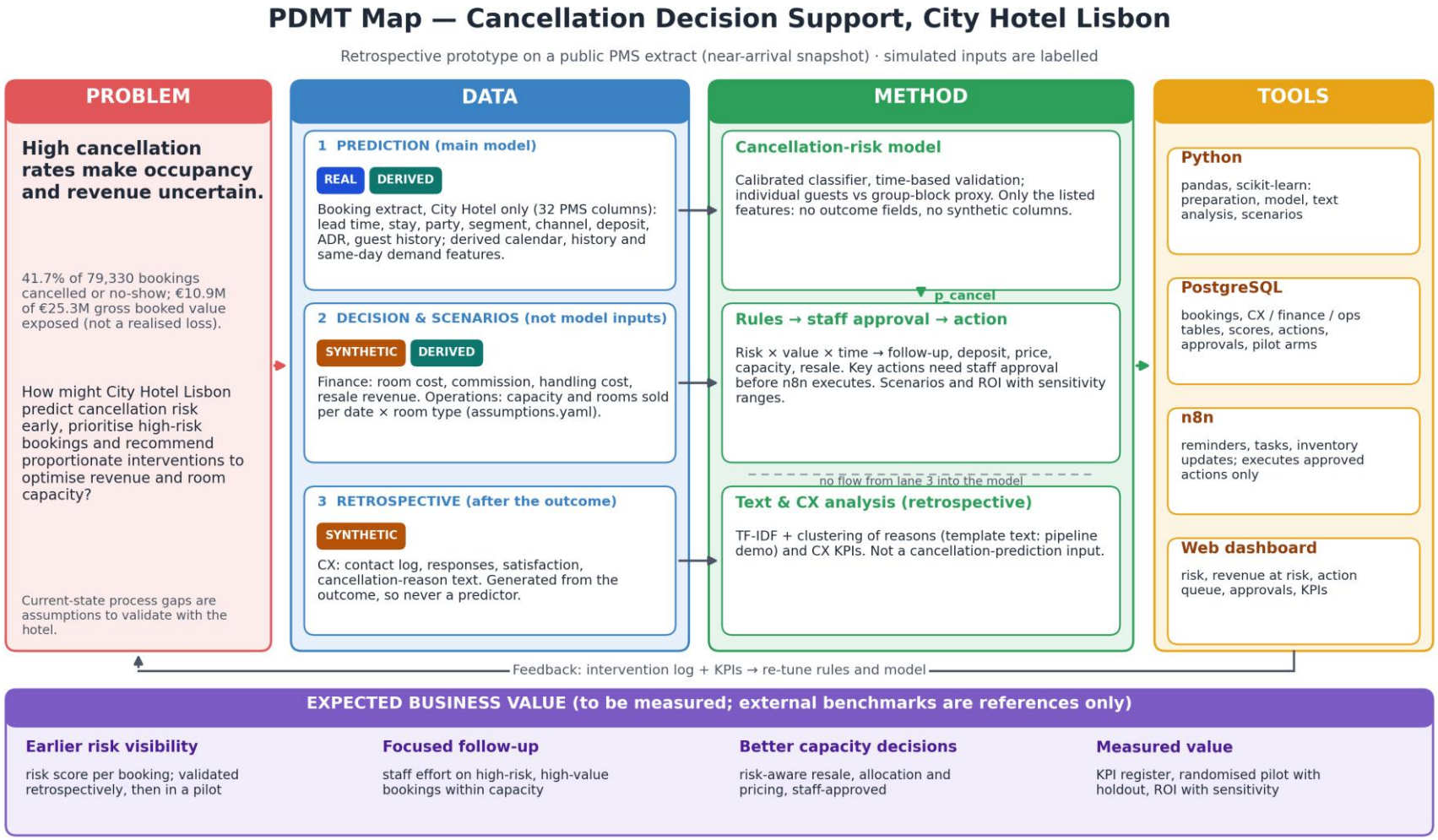
The project does not aim to eliminate cancellations or guarantee revenue gains. Targets are reference pilot goals, to be confirmed through model evaluation (see Part 3).

1.6. SCQ Framework

Situation	City Hotel Lisbon sells rooms in advance. Over July 2015 – August 2017, 41.7% of 79,330 bookings were cancelled or no-show, tying up €10.9M of €25.3M gross booked room revenue.
Complication	12.0% of cancellations (Canceled only) occur within 7 days of arrival, when resale is hardest. Risk differs sharply by segment (Groups 68.9% vs. Direct 17.3%), yet, by assumption to be validated, at-risk guests are not contacted by priority and commission and handling cost are not linked to risk. Manual review of every booking is not scalable.
Question	How might City Hotel Lisbon predict booking cancellation risk early, prioritize high-risk bookings, and recommend proportionate interventions to optimize revenue and room capacity?

> **Problem statement:** *How might City Hotel Lisbon predict booking cancellation risk early, prioritize high-risk bookings, and recommend proportionate interventions to optimize revenue and room capacity?*

2. PDMT Map


PDMT Map — Cancellation Decision Support, City Hotel Lisbon
Retrospective prototype on a public PMS extract (near-arrival snapshot) · simulated inputs are labelled
PROBLEM
High cancellation rates make occupancy and revenue uncertain.
41.7% of 79,330 bookings cancelled or no-show; €10.9M of €25.3M gross booked value exposed (not a realised loss).
How might City Hotel Lisbon predict cancellation risk early, prioritise high-risk bookings and recommend proportionate interventions to optimise revenue and room capacity?
Current-state process gaps are assumptions to validate with the hotel.
DATA
1 PREDICTION (main model)
REAL
DERIVED
Booking extract, City Hotel only (32 PMS columns): lead time, stay, party, segment, channel, deposit, ADR, guest history; derived calendar, history and same-day demand features.
2 DECISION & SCENARIOS (not model inputs)
SYNTHETIC
DERIVED
Finance: room cost, commission, handling cost, resale revenue. Operations: capacity and rooms sold per date × room type (assumptions.yaml).
3 RETROSPECTIVE (after the outcome)
SYNTHETIC
CX: contact log, responses, satisfaction, cancellation-reason text. Generated from the outcome, so never a predictor.
METHOD
Cancellation-risk model
Calibrated classifier, time-based validation; individual guests vs group-block proxy. Only the listed features: no outcome fields, no synthetic columns.
p_cancel
Rules → staff approval → action
Risk × value × time → follow-up, deposit, price, capacity, resale. Key actions need staff approval before n8n executes. Scenarios and ROI with sensitivity ranges.
no flow from lane 3 into the model
Text & CX analysis (retrospective)
TF-IDF + clustering of reasons (template text: pipeline demo) and CX KPIs. Not a cancellation-prediction input.
TOOLS
Python
pandas, scikit-learn: preparation, model, text analysis, scenarios
PostgreSQL
bookings, CX / finance / ops tables, scores, actions, approvals, pilot arms
n8n
reminders, tasks, inventory updates; executes approved actions only
Web dashboard
risk, revenue at risk, action queue, approvals, KPIs
Feedback: intervention log + KPIs → re-tune rules and model
EXPECTED BUSINESS VALUE (to be measured; external benchmarks are references only)
Earlier risk visibility
risk score per booking; validated retrospectively, then in a pilot
Focused follow-up
staff effort on high-risk, high-value bookings within capacity
Better capacity decisions
risk-aware resale, allocation and pricing, staff-approved
Measured value
KPI register, randomised pilot with holdout, ROI with sensitivity

3. Detailed Expected Business Value

Baseline: City Hotel Lisbon, Jul 2015–Aug 2017 (41.7% cancellation rate). Targets without a sound benchmark will be set from model evaluation and baseline analysis. External benchmarks are pilot references from other settings.

Business function	Expected value and target	Evaluation and data
1. Cancellation risk prediction	Provisional goal: recall ≥ 70% on the 2017 test cohort of individual bookings, i.e. detect at least 70% of cancelled or no-show bookings, with the threshold chosen on validation folds to limit false alerts.	Recall, Precision, PR-AUC and Brier score vs. a baseline model on the 2017 arrival-cohort test set (a partially explored holdout: EDA looked at 2017 labels). Data: the feature lists in config/column_groups.json (e.g. lead_time, market_segment, distribution_channel, previous_cancellations; deposit_type for group blocks). booking_changes is excluded because it accumulates until check-in or cancellation.
2. Risk-based prioritisation	Focus resources on high-risk bookings: measure the actual cancellation rate in the top 20% by risk score.	Precision@20% and Lift@20% = Precision@20% ÷ event rate of the same evaluation cohort (e.g. 34.4% for the individual 2017 test cohort), not a fixed 41.7%. Data: model risk score, is_canceled. Formulas, owners and baselines: docs/kpi_register.csv.
3. Customer experience and retention	Pilot: act on the highest-risk bookings instead of broad contact. The threshold is chosen locally on validation folds from staff capacity, intervention cost and expected benefit. The 75% threshold from churn research on repeat customers (van Leeuwen et al., 2025) is a reference only, not an established cancellation threshold.	Retention uplift (pp) = stay rate of all bookings randomly assigned to treatment − stay rate of all bookings assigned to the holdout (intention to treat, whether or not contact succeeded); also response and cancellation rates. A randomised holdout is required for a causal claim. Data: pilot contact log with a randomised holdout and is_canceled (the simulated CX table cannot estimate uplift).
4. Finance and revenue recovery	Reference: in a numerical example with one hotel and two third-party websites, overbooking raised hotel profit by	Net recovery on one contribution basis = resale revenue × (1 − commission) − room cost of

Business function	Expected value and target	Evaluation and data
	4.20% (Dong & Ling, 2015, abstract); see also Pimentel et al. (2019). A model benchmark from another setting, not an expected gain for this hotel.	resold nights + retained deposits − handling and intervention costs. The €10.9M gross figure is not net recovery. Staff-cost floor: Portugal minimum wage €505–557 per month, 2015–2017 (Hans-Böckler-Stiftung, n.d.).
5. Room inventory and operations	Reference (qualitative): updating booking limits over time under cancellation risk, rather than optimising once, improved contribution in a simulation study (Cha, 2026). The exact figure could not be verified and is not used as a target.	Resale rate = Rooms resold after cancellation ÷ Cancelled rooms eligible for resale; also unsold room-nights by arrival date. Compare operations with and without cancellation prioritisation.

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